

Markov chain

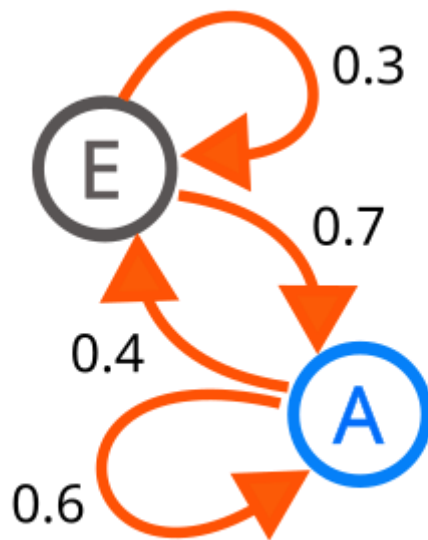
In probability theory and statistics, a **Markov chain** or **Markov process** is a *stochastic process* describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.

Informally, this may be thought of as,

"What happens next depends only on the state of affairs *now*."

Definition.

A Markov process is a stochastic process that satisfies the Markov property (sometimes characterized as "*memorylessness*").



A diagram representing a two-state Markov process - (discrete-time and countable space of states) Markov chain on a countable or finite state space. The numbers are the probability of changing from one state to another state.

History.

Andrey Markov studied Markov processes in the early 20th century, publishing his first paper on the topic in 1906.

Markov was interested in studying an extension of independent random sequences.

In his first paper on Markov chains, published in 1906, Markov showed that under certain conditions the average outcomes of the Markov chain would converge to a fixed vector of values.

Markov later used Markov chains to study the distribution of vowels in Eugene Onegin, written by Alexander Pushkin.

In his 1913 paper, Markov treated a sequence of 20,000 letters as a realization of a two-state chain of vowels and consonants.

$$\begin{bmatrix} 0,128 & 0,872 \\ 0,663 & 0,337 \end{bmatrix}$$

Based on the estimated transition probability matrix, he determined the limiting distribution of the chain [0.432 0.568].

This meant that in Russian one could expect 43.2% vowels and 56.8% consonants.

Discrete-time Markov chain

A discrete-time Markov chain is a sequence of random variables $X_1, X_2, X_3, \dots$ with the **Markov property**, namely that the probability of moving to the next state depends only on the present state and **not on the previous states**.

$$P(X_{n+1} = j | X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = P(X_{n+1} = j | P(X_n = i_n))$$

This assumption is known as Markov property.

Finite state space. The transition matrix.

If the state space is finite, the transition probability distribution for finite Markov chain with states i, j belonging to the **state space**

$S = (1, 2, 3, \dots, r)$ can be represented by a matrix, called the **transition matrix**, with the (i, j) -th element of P equal to

$$p_{ij} = P(X_{n+1} = j | P(X_n = i_n))$$

Since each row of P sums to one and all elements are non-negative, P is a **right stochastic matrix**.

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1r} \\ \vdots & \ddots & & \vdots \\ p_{r1} & p_{r2} & \cdots & p_{rr} \end{bmatrix}$$

where the following conditions are met

$$p_{ij} \geq 0 \text{ and } \sum_{j \in S} p_{ij} = 1$$

The unconditional (absolute) probability distribution

The unconditional (absolute) probability distribution of the finite state of a homogeneous Markov chain at time n is called the row vector

$$\mathbf{d}_n = [d_{n1} \ d_{n2} \ \dots \ d_{nr}]$$

Where $d_{nj} = P(X_n = j)$ for $j \in S$ and

$n \in N$ expresses steps – countable change of state

In simpler terms, it is a process for which predictions can be made regarding future outcomes based solely on its present state and—most importantly—such predictions are just as good as the ones that could be made knowing the process's full history.

$$d_{ij} \geq 0 \text{ and } \sum_{j \in S} d_{ij} = 1$$

Furthermore the unconditional probability can be calculated from the well-known formula for total probability

$$d_{nj} = \sum_{i,j \in S} d_{n-1,i} * p_{ij}$$

In the matrix notation we get the dependence of unconditional probability distributions as

$$\mathbf{d}_n = \mathbf{d}_{n-1} * \mathbf{P}$$

It is a matrix recurrence equation that, with given an initial condition $\mathbf{p}_0$, has a solution of the form

$$\mathbf{d}_n = \mathbf{d}_0 * \mathbf{P}^n$$

where $\mathbf{P}^n$ the n -th power of the transition matrix is the matrix of probabilities of transition of the chain of states in n steps.

Applications

Markov chains have many applications as statistical models of real-world processes.

They provide the basis for *general stochastic simulation* methods known as *Markov chain Monte Carlo*, which are used for simulating sampling from complex probability distributions.

Markov chains have found application in areas including:

- Bayesian statistics,
- biology, chemistry,
- economics, finance,
- information theory,
- physics,
- signal processing, music and speech processing.

Markov processes are named in honor of the Russian mathematician Andrey Markov.

Markov chain models have been used in advanced baseball analysis since 1960, although their use is still rare.

Each half-inning of a baseball game fits the Markov chain state when the number of runners and outs are considered.

During any at-bat, there are 24 possible combinations of number of outs and position of the runners.

Mark Pankin shows that Markov chain models can be used to evaluate runs created for both individual players as well as a team.^[95] He also discusses various kinds of strategies and play conditions: how Markov chain models have been used to analyze statistics for game situations such as [bunting](#) and [base stealing](#) and differences when playing on grass vs. [AstroTurf](#).